

Image Data Model (Malaria Cell Detection)

a. General Information on Dataset

Dataset Name	img_dataset (containing 'Parasitized' and 'Uninfected' subfolders)
Task	Malaria Cell Detection
Number of Classes	2
Class Labels	Parasitized : 1 , Uninfected : 0
Total Samples Loaded	19908 (9954 Parasitized, 9954 Uninfected)
Resized Sample Size	64x64
Training/Testing Split	Train: 80%, Test: 20%

b. Implementation Details

Image Preprocessing Steps	1. Resize to 64x64 2. Normalization (scale from [0,255] to [0, 1]) 3. Flatten to 1D array
Feature Scaling	StandardScaler

Hyperparameters

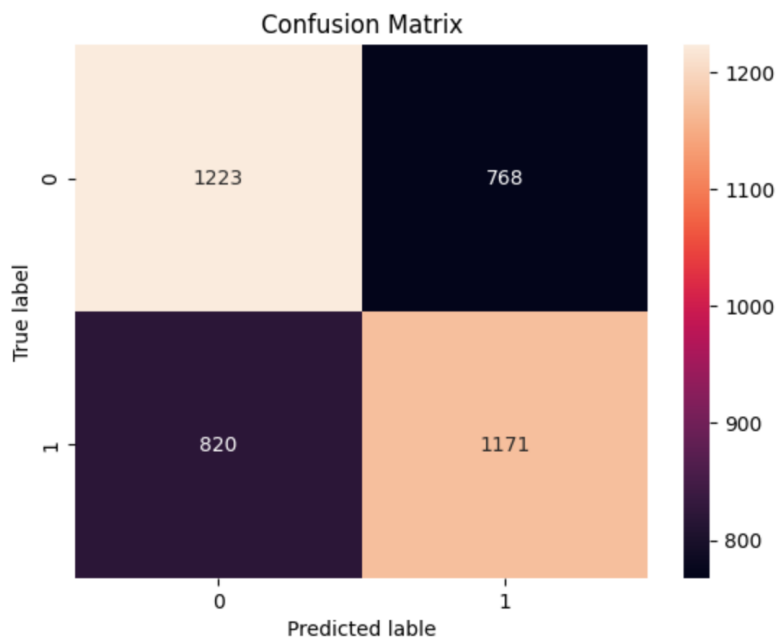
	K-Means	Logistic Regression
Hyperparameters	n_clusters=2, random_state=12	solver='saga', max_iter=500

c. Results Details Logistic Regression

Accuracy	0.6015
AUC Score	0.6368

Graphical Results

- Logistic Regression



- K-means

